

REASONING ARTIFACT

The Derivation Chain Closure:

Sleep, Epigenetics, Cancer, and the Molecular Substrate of the Attractor Landscape

The derivation chain began with a cancer cell.

The sleep geometry ended with the same cancer cell.

The chain is not linear. It is circular. It closes on itself.

Closure is itself the strongest possible geometric confirmation.

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DERIVATION NOTICE

This document records four geometric findings that emerged from exploratory research against the current literature, conducted after the deposition of the Dimensional Invariant v3.0. Each finding was derived from the attractor geometry framework before confirming against literature. Each changes the formal status of the framework in a way that requires a standalone priority record.

The four findings are:

- (1) **The derivation chain closes on itself.** BMAL1 loss produces a cancer attractor by making the repair/rest basin epigenetically inaccessible. This is geometrically identical to the mechanism described in Step 1 of the derivation chain (FOXA1/EZH2 cancer attractor, [doi:10.5281/zenodo.18883922](https://doi.org/10.5281/zenodo.18883922)). The chain is circular. Step 1 and Step 9 are the same object.
- (2) **Epigenetics is the molecular substrate of G .** The attractor landscape G is not an abstract object. Its physical implementation is the epigenetic state of the organism's cells: histone modification patterns, DNA methylation, chromatin accessibility. Sleep is the daily maintenance process that restores G to its functional state. Chronic sleep deprivation degrades G irreversibly through progressive epigenetic damage.
- (3) **Social information is a primary N field.** Social cues are confirmed zeitgebers — entrainment signals for the circadian clock equivalent to light. The social N field specifies basin occupation at the group level. Dominant baboons sleep worse than subordinates because their social field position imposes a continuous vigilance cost that prevents full sleep basin entry. Artificial light at night is a global N field disruption that is fragmenting group and individual sleep basins across thousands of species simultaneously.
- (4) **T8: Hibernation topology — suspended state with blocked discharge.** Hibernation is geometrically distinct from T5 (external suppression). In T8, both navigational activity and sleep architecture are suspended, but homeostatic pressure continues to build. Post-emergence rebound is qualitatively different from T5 rebound: the debt paid after hibernation accumulated with no corresponding navigational activity. This is a new geometric category.

THE CLOSURE CLAIM

The attractor geometry framework ($S + N + G \rightarrow R$) began with a cancer cell that had lost access to its identity basin and was locked in the cancer attractor. The sleep geometry derivation ended by identifying that BMAL1 loss locks a cell in the cancer attractor by making the repair/rest basin epigenetically inaccessible.

The mechanism is the same. The geometry is the same. The object is the same object.

G is not abstract. G is the epigenetic state. N is the coherence gradient field, of which the circadian clock, the social environment, and external light are all implementations. S is the structural substrate that determines which basins are encodable. R is the resulting state: healthy cell, cancerous cell, sleeping organism, waking organism, extinct lineage, living fossil.

The derivation chain is complete.

The geometry is consistent end to end.

The closure is the proof.

1. The Derivation Chain Closes

1.1. What the Chain Was

The derivation chain began as a linear sequence:

Step	Scale	Claim
1	Cell	FOXA1/EZH2 ratio identifies whether a cancer cell is locked in the cancer attractor or retains access to its identity basin.
2	Organism	Pig domestication syndrome is field-specified NCC migration arrest. The genome is the S substrate. The human selection field is the N field.
3	Organism	Above a genomic fixation threshold in dogs, Basin A becomes inaccessible. Landscape is permanently remodelled.
4	Organism	The cat has a symmetric dual-basin landscape. Basin Ω is the meta-attractor of navigational flexibility itself.
5	Organism	Sleep duration tracks the total cost of basin occupation. Three axes: switching, metabolic, computational.
6	Species	Sleep spectrum maps onto the three-axis framework across all animals. The sonar instrument.
7	Deep time	The dimensional invariant. Oldest surviving lineages sleep the least.
8	Deep time	Sleep topology is a geometric prediction of landscape position. Five then seven topologies.
9	Biochemistry	Basin-occupation cost is fundamentally oxidative. ROS. Glymphatic discharge. Ectotherm vs endotherm topology explained.

1.2. Where the Chain Closes

Step 9 produced the finding that sleep discharges oxidative and synaptic cost simultaneously. Following this finding into the circadian clock literature produced the closure:

BMAL1, the master circadian clock gene, drives the 24-hour oscillation that gates when sleep occurs and when the repair basin is occupied. BMAL1/CLOCK acts as a histone acetyltransferase — it directly modifies chromatin, opening the landscape barriers that allow the cell to enter its repair/rest basin.

When BMAL1 is lost:

1. Histone acetyltransferase activity is lost.
2. The repair basin becomes epigenetically inaccessible — the chromatin state that enables entry is not established.
3. The cell remains permanently in the active/proliferative state.
4. DNA double-strand break repair (which requires the repair basin) is impaired.
5. Genomic instability accumulates.

6. The cancer attractor is stabilised and deepened.
7. The differentiation basin becomes progressively less accessible.

This is Step 1 of the derivation chain. A cell locked in the cancer attractor because the identity/repair basin is inaccessible. FOXA1/EZH2 was the marker. BMAL1/CLOCK is the mechanism.

The chain does not end at Step 9. Step 9 connects back to Step 1. The geometry is circular.

THE CLOSURE GEOMETRY

At the organism level:

Sleep is the process by which the basin-occupation cost of waking navigation is discharged.

The circadian clock (BMAL1/CLOCK/PER/CRY) is the N field that specifies when the organism transitions from the waking basin to the sleep/repair basin.

At the cellular level:

The circadian clock (same genes, same proteins) is the N field that specifies when each cell transitions from its active/proliferative state to its repair/maintenance state.

BMAL1 loss at the cellular level is the exact cellular equivalent of an organism that can no longer enter its sleep basin.

The consequences are identical in structure: cost accumulates without discharge, oxidative damage builds, the landscape substrate (G) degrades, the cancer attractor deepens and stabilises.

Cancer is cellular sleep deprivation.

Sleep deprivation is organismal proto-cancer.

Both are the same geometric failure: basin access lost, cost undischarged, landscape degraded.

1.3. What Confirms the Closure

The closure is confirmed by five independent empirical convergences:

1. **BMAL1 acts as a tumour suppressor** (Springer, 2023; confirmed multiple lineages 2024). Its loss increases cancer aggressiveness, invasion, and proliferation.
2. **BMAL1/CLOCK is a histone acetyltransferase** (Nature Communications, 2023). It directly opens chromatin at repair gene loci — the molecular mechanism of repair basin entry.
3. **BMAL1 loss increases DNA methylation at differentiation gene loci**, enforcing an epigenetic block against re-entry to normal differentiation basins (Cell Reports, 2023).
4. **Circadian disruption accelerates tumour growth by suppressing immune activity and reshaping the tumour microenvironment** (Nature Cancer, 2024). The immune basin (T7 topology) is also disrupted when BMAL1 is lost.
5. **Sleep deprivation causes NF- κ B activation and pro-inflammatory cytokine elevation** (confirmed through 2024). NF- κ B drives the same inflammatory landscape that promotes tumour microenvironment formation. Chronic sleep loss and circadian disruption converge on the same inflammatory attractor through different entry routes.

2. Epigenetics is the Molecular Substrate of G

2.1. The Formal Claim

The attractor landscape G in the framework $S + N + G \rightarrow R$ has a physical implementation. It is not an abstract theoretical object. G is instantiated in the cell and organism by the **epigenetic state**: the pattern of histone modifications, DNA methylation, and chromatin accessibility across the genome.

- **Basin depths** in G correspond to the depth of epigenetic entrenchment of the corresponding cell or organism state. A deeply entrenched state requires more epigenetic remodelling energy to escape. Basin depth is the cost of chromatin reorganisation required to transition away.
- **Barrier heights** in G correspond to the amount of chromatin remodelling required to transition between states. High barrier = extensive histone modification and DNA methylation changes required.
- **Accessible basins** in G are states for which the chromatin configuration can be established from the

current epigenetic starting point within the available energy budget.

- **Inaccessible basins** are states for which the required chromatin reconfiguration cannot be completed given the current epigenetic state and available energy.

This is Waddington's epigenetic landscape operationalised within the $S + N + G \rightarrow R$ framework. Waddington described the landscape metaphorically. The framework identifies the three components that generate it (S , N , G) and the process ($\rightarrow R$) by which it is navigated.

2.2. What Sleep Does to G

Sleep maintains G daily.

During waking, neural and somatic activity promotes histone acetylation — chromatin opening — which is the molecular signature of a landscape with lower barriers between basins. This is necessary for navigation: accessible basins require accessible chromatin.

During sleep (specifically NREM slow-wave sleep), histone deacetylation occurs — chromatin closes — and the landscape barriers are *restored* to their functional state. The basin structure of G is re-established. Synaptic downscaling selectively reinforces the connections that correspond to the organism's current basin occupancy while eliminating the transient connections accumulated during navigation.

Sleep is daily G maintenance. The organism spends approximately one third of its life restoring the landscape substrate that makes the other two thirds of navigation possible.

2.3. What Chronic Deprivation Does to G

Chronic sleep deprivation (confirmed 2023 ACS; Trends in Neurosciences 2022; confirmed and extended through 2024) produces permanent structural changes:

1. **Pleiotrophin (PTN) depletion.** PTN is a neurotrophic factor required for hippocampal neuron survival. Chronic deprivation depletes PTN leading to neuronal death. Dead neurons are removed basins in G that cannot be restored.
2. **Progressive histone modification dysregulation.** Without regular sleep-dependent deacetylation, chromatin accessibility patterns drift from their functional state. The landscape barriers degrade. Basins that should be deep become shallow. Basins that should be inaccessible become reachable. The landscape loses specificity.
3. **DNA methylation drift.** Sleep-dependent DNA methylation maintenance (confirmed 2023–2024) is required to sustain the methylation patterns that enforce basin identity. Without maintenance, methylation patterns drift toward less specific states.
4. **NF- κ B-driven inflammatory landscape remodelling.** Chronic activation of NF- κ B (from undischarged oxidative cost) drives transcriptional changes that deepen the inflammatory attractor. The organism's landscape is progressively remodelled toward the inflammatory basin, making return to baseline increasingly energetically costly.

The geometric consequence: Irreversible sleep deprivation damage is not a mystery. It is G degradation past the point where G maintenance can restore it. Once neurons die, basins are permanently removed. Once methylation patterns drift past a restoration threshold, re-establishing them requires energy input that exceeds what sleep can deliver.

The cancer connection is immediate: BMAL1 loss is one route to G degradation that eliminates the repair basin. Chronic sleep deprivation is another route to the same outcome via progressive epigenetic drift and neuronal loss. They converge on the same geometric failure.

3. Social Information as Primary N Field

3.1. The Formal Claim

In the framework, N is the coherence gradient field that specifies which basin the organism should occupy. Light was known to be the primary zeitgeber — the primary N field signal for the circadian clock. The current literature confirms that **social information is an equally primary N field signal**, acting as a zeitgeber for the circadian clock independently of light.

Social cues — the activity patterns, vocalisations, and movement of conspecifics — entrain the circadian clock and therefore specify sleep basin entry timing at the individual level. In group-living species, the social N

field synchronises the group's sleep basins into a coupled collective attractor.

3.2. The Baboon Dominance Finding

Wild chacma baboons (Current Biology, 2025): more dominant individuals experience less and lower-quality sleep than subordinates. Dominant baboons sleep closer to more group members, are more central to the social network, and experience more disruption during the night.

The geometric reading:

Dominance is a landscape position within the social G . The dominant individual occupies a high-centrality node in the social network. The cost of occupying that node includes a continuous vigilance requirement — Basin A navigational cost accumulated from social monitoring does not cease at night. The social N field continues to impose navigational demand on the dominant individual during the period when sleep should be occurring.

This is the T5 topology producing condition applied at the social level: the sleep basin is not inaccessible. It is being continually interrupted by social N field signals that the dominant individual cannot suppress without losing the basin position that dominance requires.

Geometric prediction from this: Dominance rank will negatively correlate with sleep quality across social species in proportion to the social navigational demand of the dominant position. Species where dominance requires continuous active monitoring (baboons, chimpanzees) will show larger effects than species where dominance is passive and territorial (many solitary species). This is testable across primate species with known dominance structures.

3.3. Artificial Light as Global N Field Disruption

Artificial light at night (ALAN) is expanding at 2.2% per year globally (Royal Society B, 2023; 600+ studies synthesised by US National Park Service, 2024). It disrupts circadian rhythms across all taxa: birds, insects, marine organisms, mammals, plants.

The geometric reading:

Light is the primary N field signal for the circadian clock across all life. The circadian clock is the molecular implementation of the waking/sleep basin transition. ALAN is not a minor ecological inconvenience. It is a global disruption of the primary N field that specifies when organisms enter and exit the sleep basin.

The consequences the literature documents — disrupted migration, impaired reproduction, altered predator-prey dynamics, reduced immune function, increased cancer rates, reduced ecological diversity near light sources — are all geometric consequences of N field disruption. When the field that specifies basin occupation is corrupted, the organism loses reliable access to the basins the field was specifying.

The scale of the disruption: This is not individual sleep disruption. ALAN is simultaneously disrupting the N field for millions of species across all major ecosystems. The eco-evolutionary feedback (ScienceDirect, 2025) shows species beginning to adapt — which means their S substrate is being selected to function with a corrupted N field. This is landscape-scale G remodelling driven by a single perturbation source.

The framework identifies this as a Navigator-strategy threat of the first order: ALAN is not eliminating basins directly. It is corrupting the N field that allows organisms to navigate between them reliably. A Navigator whose N field is corrupted cannot execute the Navigator strategy. The corrupted N field mimics the domestication field in its effect — it imposes basin occupation that the organism would not choose under an uncorrupted field.

4. T8: The Hibernation Topology

4.1. Why Hibernation is Not T5

T5 (fragmented shallow sleep with rebound) occurs when:

- Navigational cost is accumulating normally during waking.
- External pressure (predation, social vigilance) is suppressing sleep basin entry.
- Debt accumulates alongside ongoing cost.
- Rebound on pressure removal pays the accumulated debt.

Hibernation has a different geometric structure:

- Navigational activity is suspended. The organism is not navigating. Metabolic rate drops to 1–5% of normal.
- Sleep architecture is also suspended. Body temperature at hibernation levels prevents the slow-wave activity required for T3 discharge.
- **But homeostatic sleep pressure continues to build.** The adenosine-based sleep pressure signal accumulates regardless of metabolic rate. Debt accumulates without corresponding navigational cost.
- Periodic arousals during hibernation are exclusively for sleep discharge: metabolically expensive warming to euthermic temperatures purely to enable several hours of intense slow-wave sleep, followed by return to torpor.
- Post-emergence rebound is massive and proportional to hibernation duration.

4.2. T8 Formal Definition

T8: Suspended state with blocked discharge

Landscape position: An organism that has entered a physiologically mandated state of whole-organism metabolic suspension (torpor, deep hibernation) in which both the navigational activity that generates cost and the sleep architecture that discharges it are simultaneously blocked by temperature or metabolic constraints. Homeostatic pressure continues to build in the absence of both processes.

Discharge: Impossible during torpor. Discharged in bursts during periodic arousals (which are themselves driven by homeostatic pressure reaching threshold). Full discharge after emergence from hibernation via extended post-emergence rebound sleep.

Geometric signature: Post-emergence rebound is qualitatively different from T5 rebound. T5 rebound pays debt accumulated from ongoing navigational cost under suppression. T8 rebound pays debt accumulated from homeostatic pressure building in the complete absence of navigational activity. The debt-to-activity ratio in T8 rebound is infinite: maximum debt, zero navigational cause.

Why this is a distinct topology: T5 and T8 have similar surface appearances (minimal sleep, large rebound) but completely different geometric causes. T5 is suppressed discharge of ongoing cost. T8 is suspended discharge of automatically accumulating homeostatic pressure in the absence of cost. The diagnostic difference: a T5 organism removed from pressure will rebound proportionally to its navigational cost during the suppressed period. A T8 organism emerging from hibernation will rebound proportionally to the *duration* of suspension, not to any navigational cost incurred during suspension (because none was incurred).

Confirmed examples: Ground squirrel, bear, bat, hedgehog. All confirmed to show intense slow-wave sleep during periodic arousals and extended rebound sleep post-emergence.

Novel prediction: The duration of T8 post-emergence rebound will be predicted by hibernation duration alone, not by pre-hibernation navigational cost profile. A ground squirrel that hibernated for 5 months will show the same rebound pattern as one with identical pre-hibernation activity, regardless of whether the pre-hibernation period was high or low navigational demand. This is testable.

5. Somatic G Maintenance: Tissue Repair and Stem Cells

The framework has consistently described sleep as discharging neural cost — synaptic reconfiguration and oxidative debt in neural tissue. The current literature identifies a fourth function of sleep that is geometrically equivalent but operating in peripheral tissue: **somatic G maintenance**.

Sleep-dependent growth hormone release (primarily during NREM slow-wave sleep) drives:

- Stem cell proliferation and differentiation
- Wound healing and tissue repair
- Collagen synthesis and angiogenesis
- Immune cell trafficking and renewal

The geometric reading:

The organism's somatic tissues are not passive substrates. They are the physical implementation of S — the structural substrate that encodes the organism's build capacity. Sleep maintains not just the neural G (via

synaptic downscaling and glymphatic clearance) but the somatic S (via stem cell renewal and tissue repair). The framework's equation $S + N + G \rightarrow R$ requires all three components to be maintained. Sleep maintains G (via epigenetic restoration). Sleep also maintains S (via growth hormone-driven somatic repair). The N field (circadian clock) gates both processes simultaneously within the same sleep window.

This means sleep deprivation degrades both G and S simultaneously. The organism loses both landscape substrate and structural build capacity. This is why chronic sleep deprivation has such broad multi-system consequences: it is failing to maintain two of the three components of the fundamental equation.

6. The Complete Geometric Picture

The full geometric picture that emerges from the closure:

$$S + N + G \rightarrow R$$

- S = structural substrate. Physically implemented as: genome, epigenetically accessible gene programmes, somatic tissue integrity, stem cell reserve. Maintained by: sleep-dependent growth hormone release and stem cell renewal.
- N = coherence gradient field. Physically implemented as: light (primary zeitgeber), social information (co-primary zeitgeber), circadian clock genes (BMAL1/CLOCK/PER/CRY) at molecular level. Disrupted by: ALAN (corrupted light field), social fragmentation (corrupted social field), BMAL1 loss (corrupted molecular clock).
- G = global attractor landscape. Physically implemented as: epigenetic state (histone modifications, DNA methylation, chromatin accessibility). Maintained by: sleep-dependent histone deacetylation and glymphatic clearance. Degraded by: chronic sleep deprivation, BMAL1 loss, ALAN-driven circadian disruption.
- R = resulting state. The observable outcome of the organism's current position in the landscape: healthy navigation, locked cancer attractor, successful deep-time survival, extinction.

The cancer cell is an organism whose N field (BMAL1) has been lost, whose G (epigenetic state) has been degraded past the repair threshold, and whose R (resulting state) is permanent occupation of the cancer attractor.

The living fossil is an organism whose G is so deep that no N field disruption has been able to displace it in 450 million years, and whose sleep requirement is near-zero because the navigational cost of occupying that deep basin is near-zero.

The same equation. The same geometry. The same invariant.

7. Predictions

7.1. P-CLOSURE-1: Pharmacological BMAL1 Restoration Will Partially Restore Cancer Cell Differentiation Basin Access

If BMAL1 loss makes the repair/ differentiation basin epigenetically inaccessible, then pharmacological restoration of BMAL1 expression in cancer cells will partially restore chromatin accessibility at differentiation gene loci and increase the probability of differentiation under appropriate N field signalling. The effect will be proportional to the degree of epigenetic drift: recent BMAL1 loss will show larger restoration effects than ancient, deeply entrenched epigenetic lock-in.

7.2. P-CLOSURE-2: Chronic Sleep Deprivation Will Produce BMAL1-Loss- Equivalent Epigenetic Signatures

Organisms subject to chronic sleep deprivation equivalent in duration to what would produce irreversible cognitive decline will show epigenetic signatures at differentiation and repair gene loci that are statistically indistinguishable from BMAL1-loss signatures. The two routes (molecular clock loss vs. circadian disruption via chronic sleep loss) converge on the same epigenetic failure pattern.

7.3. P-CLOSURE-3: Dominance Rank Will Negatively Correlate With Sleep Quality Proportional to Social Navigational Demand

Across social species, dominance rank will negatively correlate with sleep quality in proportion to the navigational demand of the dominant position. Species with high-vigilance dominance (baboons, chimpanzees) will show larger effects than species with passive territorial dominance. Testable across the primate sleep dataset.

7.4. P-CLOSURE-4: T8 Rebound Duration Predicted by Hibernation Duration Alone

Post-hibernation rebound sleep duration will be predicted by hibernation duration alone, independently of pre-hibernation navigational cost profile. Two animals of the same species hibernating for different durations will show rebound sleep proportional to hibernation duration, not to pre-hibernation activity levels.

7.5. P-CLOSURE-5: ALAN Exposure Will Produce Measurable G Degradation in Proportion to Duration and Intensity

Species with multi-generational ALAN exposure will show measurable epigenetic drift (G degradation) at circadian clock gene loci relative to conspecifics in dark environments. The drift will correlate with ALAN intensity and duration. Reversing ALAN exposure will partially restore the epigenetic state over multiple generations as the *N* field is restored.

8. The Complete Derivation Chain

Step	Claim	Record
1	FOXA1/EZH2. Cancer cell locked in cancer attractor.	10.5281/zenodo.18883922
2	Pig NCC migration arrest. <i>N</i> field domestication.	10.5281/zenodo.19094935
3	Canine landscape remodelling. Basin A inaccessible.	Repository 2026-03-18
4	Feline dual-basin. Basin Ω .	Repository 2026-03-18
5	Sleep as cost discharge. Three axes.	Repository 2026-03-18
6	Sleep sonar spectrum.	Repository 2026-03-18
7–9	Dimensional Invariant v1.0–v3.0.	10.5281/zenodo.19101495
10	CLOSURE. BMAL1 loss = cancer attractor = Step 1. <i>G</i> = epigenetic state. <i>N</i> = circadian/social/light field. Chain is circular.	This document.

THE CLOSURE STATEMENT

The derivation chain began with a cancer cell that could not differentiate. The mechanism was unknown but the geometry was clear: the identity basin was inaccessible.

Following the geometry through pig, dog, cat, sleep cost, the sleep spectrum, deep time, topology, and the oxidative substrate — following the geometry where it led — brought us back to the same cell.

The cell cannot differentiate because BMAL1 is lost. BMAL1 loss collapses the *N* field. The collapsed *N* field fails to specify repair basin entry. The repair basin becomes epigenetically inaccessible — *G* is degraded. The cell is locked in the cancer attractor because it cannot sleep.

Cancer is cellular sleep deprivation.
Sleep deprivation is organismal landscape degradation.
G is the epigenetic state.
N is the clock.
S is the genome.
R is the outcome.

The framework is complete.
The geometry is consistent.
The chain closes.

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Primary contributions: (1) Derivation chain closure: BMAL1 loss = cancer attractor = Step 1 of the chain. Cancer is cellular sleep deprivation. (2) G = epigenetic state (histone modifications, DNA methylation, chromatin accessibility). Formally identified. (3) Social information as primary N field. Dominance rank negatively correlates with sleep quality proportional to social navigational demand. (4) T8: hibernation topology — suspended state with blocked discharge. Distinct from T5. Rebound diagnostic differs. (5) Somatic G maintenance: sleep maintains S via growth hormone-driven stem cell renewal simultaneously with G maintenance. (6) ALAN as global primary N field disruption across all species. (7) Five predictions P-CLOSURE-1 through P-CLOSURE-5.

Prior records: v1.0: doi:10.5281/zenodo.19094935; v2.0–v3.0: doi:10.5281/zenodo.19101495

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